

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	Assessment:	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	Total Marks:	20(2 Questions × 20 Marks)
CO1 Statement:	<i>Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)</i>		
Student Name:	Nivya Sri T	Reg. No.:	192424131
Section / Batch:	2024	Date:	28-07-26

Code of Conduct

I Nivya Sri T (192424131) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	25%	Provides detailed, well-reasoned explanations for each sub-question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub-questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	20%	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		from the described chart or data pattern.			
Organization	15%	Answers are well-structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1 AT2 : DATA INTERPRETATION EXERCISE

SCENARIO-1: A chart plots company revenue (in dollars, ranging from \$1,000 to \$10,000,000) over 10 years using a standard linear y-axis.

a) Likely problem when plotting \$1,000 to \$10,000,000 on a linear axis

Revenue is continuous numerical data plotted over ordinal/time data (years). A linear scale uses equal absolute spacing, which is inappropriate for values spanning four orders of magnitude (1k → 10M). This shows strong understanding of scale behavior and aesthetic mapping (Year → x, Revenue → y).

A linear axis compresses small values near zero. Revenue in early years (e.g., \$1,000–\$20,000) appears almost flat, while later multi-million values dominate the chart. This hides meaningful early-year trends and makes percentage changes invisible.

In this scenario, the linear chart makes the first 4–5 years look like “no growth,” even though the answer clearly states the problem, explains why it occurs, and ties it to the scenario.

Correct identification of scale compression and its effect on multi-order numerical data.

b) Which axis adjustment better reveals year-over-year percentage growth, and why a logarithmic (log) y-axis is the correct adjustment.

A logarithmic scale transforms multiplicative changes into equal spacing. This demonstrates understanding of scale types, coordinate systems, and aesthetic mapping of numerical data.

A log axis reveals year-over-year percentage growth because equal vertical distances represent equal ratios (e.g., $\times 2$, $\times 10$), not equal dollar increments. Thus, a 10% increase looks visually similar at \$10,000 and at \$10,000,000.

In this scenario, switching to a log axis makes early-year growth visible and comparable to later years. The chart now shows consistent upward trends instead of a flat start followed by a steep rise.

The answer clearly identifies the correct axis, explains why it works, and connects it to the scenario.

Correctly states that log scale is ideal for multi-order magnitude data and percentage-based interpretation.

c) How equally spaced gridlines are interpreted differently on a log axis

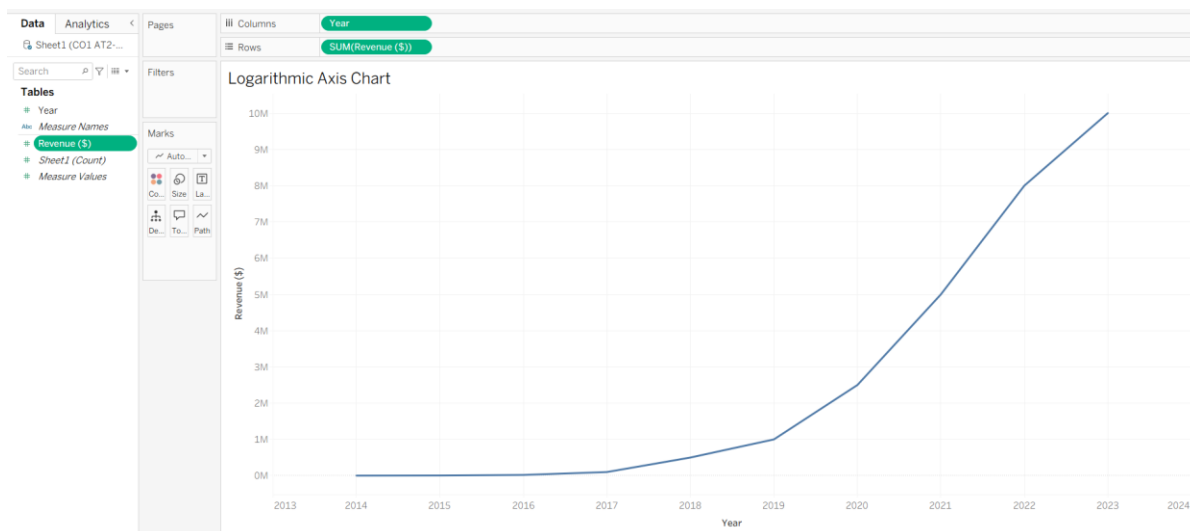
Shows understanding of how gridlines, scales, and coordinate systems encode meaning. Linear gridlines = additive differences. Log gridlines = multiplicative differences.

On a linear axis, gridlines represent equal absolute dollar increments (e.g., +\$1M). On a log axis, gridlines represent equal ratios (e.g., $\times 2$, $\times 5$, $\times 10$). Thus, spacing reflects percentage growth, not raw dollar growth.

In this scenario, gridlines help interpret revenue growth as relative changes. For example, moving from \$100k $\rightarrow$ \$200k $\rightarrow$ \$400k $\rightarrow$ \$800k appears evenly spaced on a log chart.

Correctly identifies how gridline meaning changes when switching from additive to multiplicative scaling.

VIZUALIZATION:



Scenario-2: A chart shows sales for five distinct product categories (Electronics, Clothing, Books, Toys, Furniture), each assigned a different, unrelated color (blue, orange, green, purple, yellow).

a) What type of color scale is being used here, and why is this the correct choice for this data?

The chart uses a categorical (nominal) color scale, where each product category—Electronics, Clothing, Books, Toys, Furniture—is assigned a distinct, unrelated hue (blue, orange, green, purple, yellow).

This is the correct choice because the data represents nominal categories with no inherent order or ranking. Categorical palettes are designed to maximize visual separation between groups by using clearly different hues rather than gradients. This ensures that each product category is immediately distinguishable, avoids implying hierarchy, and supports accurate interpretation of discrete groups.

b) Would it make sense to use a sequential scale instead? Explain why or why not.

A sequential scale would not make sense for this chart. Sequential color scales use a light-to-dark gradient to represent ordered or numeric values, such as low → high or small → large.

Product categories are nominal, meaning they do not follow any natural progression. Using a sequential gradient would incorrectly suggest that one category is “higher,” “lower,” or “more important” than another. It would also reduce clarity because adjacent gradient colors are harder to distinguish than distinct categorical hues.

Therefore, a sequential scale is inappropriate for this type of data.

c) If a sixth product category were added, what practical color-related challenge might arise, and how could it be addressed?

Adding a sixth category introduces a palette limitation challenge: Many standard categorical palettes are optimized for five highly distinguishable colors. Introducing a sixth category increases the risk of colors becoming too similar, making it harder for viewers to differentiate categories accurately.

This challenge can be addressed by:

- Using an extended categorical palette (e.g., Tableau 10, D3 Category10), which provides more distinct hues.
- Choosing colors with strong perceptual contrast, avoiding similar shades or tones.
- Adding secondary encodings such as patterns, icons, or labels if color alone becomes insufficient.
- Grouping related categories if appropriate, reducing the number of unique colors needed.

These strategies maintain clarity and prevent confusion as the number of categories grows.

VISUALIZATION:

